

Interactive CPU Scheduling Simulator with AI-Powered Flaw Analysis

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Abstract—CPU scheduling is a fundamental component of modern operating systems, directly influencing process throughput, response time, and overall system efficiency. This paper presents an interactive, web-based CPU scheduling simulator that supports eight scheduling algorithms—First Come First Serve (FCFS), Shortest Job First (SJF), Shortest Remaining Time First (SRTF), Priority Scheduling (non-preemptive, preemptive, and with aging), Round Robin (RR), Multi-Level Queue (MLQ), and Multi-Level Feedback Queue (MLFQ). The simulator provides real-time Gantt chart visualization, per-process statistics, and a side-by-side algorithm comparison mode. A novel contribution of this work is the integration of an AI-powered flaw analysis engine that automatically detects scheduling anomalies such as the convoy effect, starvation, excessive context switching, queue starvation in multi-level schedulers, and boost dependency in MLFQ. The system is built using Next.js, TypeScript, and Zustand for state management, and leverages large language models via the OpenAI API for intelligent analysis. Our results demonstrate that the simulator serves as an effective educational tool for understanding the trade-offs inherent in CPU scheduling algorithms.

Index Terms—CPU Scheduling, Operating Systems, Multi-Level Feedback Queue, AI-Powered Analysis, Interactive Simulation

I. INTRODUCTION

CPU scheduling is one of the most critical functions performed by an operating system kernel. The scheduler decides which process in the ready queue gets access to the CPU and for how long, directly affecting key performance metrics such as throughput, turnaround time, waiting time, and response time [1].

Traditional textbooks present scheduling algorithms in a static, theoretical manner—students compute Gantt charts by hand and struggle to visualize the dynamic behavior of preemptive algorithms like SRTF or MLFQ. While several scheduling simulators exist [2], [4], they typically support only basic algorithms and lack real-time interactivity or intelligent feedback.

This paper makes the following contributions:

- A fully interactive, web-based simulator supporting eight CPU scheduling algorithms, including MLQ and MLFQ with configurable queue parameters.
- Real-time visualization through animated Gantt charts, ready queues, I/O queues, and CPU core animations.

- An AI-powered flaw analysis engine that automatically detects and explains scheduling anomalies upon simulation completion.
- A comparison mode enabling side-by-side evaluation of multiple algorithms on identical workloads.

The remainder of this paper is organized as follows: Section II reviews related work. Section III describes the system architecture and methodology. Section IV presents results and discussion. Section V concludes with future directions.

II. RELATED WORK

Several CPU scheduling simulators have been developed for educational purposes. Jones et al. [4] developed a desktop-based simulator supporting FCFS, SJF, and RR but lacked support for multi-level scheduling algorithms. The initial framework for operating system simulation was established in [1], while more recent advancements in interactive OS education were explored in [2].

Web-based approaches have gained traction due to their accessibility. However, existing web simulators [5] typically do not support MLFQ with configurable priority boost or provide AI-driven analysis of simulation results. Our work bridges this gap by combining comprehensive algorithm coverage with intelligent, automated flaw detection.

III. METHODOLOGY

A. System Architecture

The simulator is built as a single-page application using the Next.js framework with TypeScript for type safety. State management is handled by Zustand, a lightweight reactive store. The architecture follows a clean separation of concerns:

- **Scheduling Engine** (`lib/schedulers/`): Contains individual scheduler implementations that conform to a common `Scheduler` interface.
- **State Store** (`lib/store.ts`): Centralized application state including process lists, simulation state, and play-back controls.
- **Analysis Module** (`lib/analysis/`): Prepares simulation data and communicates with the OpenAI API for flaw detection.

- **UI Components** (`components/`): React components for visualization, including Gantt charts, queue displays, and statistics panels.

B. Scheduling Algorithms

The simulator implements the following algorithms:

TABLE I
SUPPORTED SCHEDULING ALGORITHMS

Algorithm	Selection Criteria	Preemptive
FCFS	Arrival order	No
SJF	Shortest burst	No
SRTF	Shortest remaining	Yes
Priority	Lowest priority #	No
Priority-P	Lowest priority #	Yes
Aging	Effective priority	Yes
Round Robin	FCFS + quantum	Yes
MLQ	Highest queue	Yes
MLFQ	Adaptive queues	Yes

C. Simulation Tick Engine

Each simulation proceeds in discrete time units (“ticks”). The `executeTick()` function follows an 8-step pipeline per tick:

- 1) Admit newly arrived processes to the ready queue.
- 2) Decrement I/O timers; move completed I/O processes back to ready.
- 3) Check preemption conditions (for preemptive algorithms).
- 4) Select the next process via `selectNextProcess()`.
- 5) Execute one CPU time unit on the running process.
- 6) Handle burst completion (move to I/O or terminate).
- 7) Increment waiting time for all ready-queue processes.
- 8) Record the Gantt chart entry.

MLQ and MLFQ use dedicated tick functions (`mlqTick()`, `mlfqTick()`) due to their multi-queue nature.

D. Multi-Level Feedback Queue (MLFQ)

The MLFQ implementation follows the five canonical rules [3]:

- **Rule 1:** New processes enter the highest-priority queue (Q0).
- **Rule 2:** Processes that do not exhaust their time quantum remain at the same priority level.
- **Rule 3:** Processes that consume their full quantum are demoted.
- **Rule 4:** I/O-bound processes retain their priority level.
- **Rule 5/6:** A periodic priority boost moves all processes to Q0 to prevent starvation.

The demotion condition is checked as:

$$\text{demote}(P) = \begin{cases} \text{true} & \text{if } t_{\text{used}} \geq Q_{\text{quantum}} \\ \text{false} & \text{otherwise} \end{cases} \quad (1)$$

where t_{used} is the CPU time consumed in the current queue and Q_{quantum} is the time quantum of that queue level.

E. Priority Aging

To prevent starvation in priority scheduling, the simulator implements aging. The effective priority of a process P is computed as:

$$\text{priority}_{\text{eff}}(P) = \text{priority}(P) - \left\lfloor \frac{w(P)}{A} \right\rfloor \quad (2)$$

where $w(P)$ is the accumulated waiting time and A is the configurable aging interval.

F. AI-Powered Flaw Analysis

Upon simulation completion, the system automatically invokes the analysis engine. The pipeline operates as follows:

- 1) **Data Preparation:** Process statistics, Gantt chart entries, per-queue metrics, and algorithm configuration are serialized into a structured prompt.
- 2) **LLM Invocation:** The prompt is sent to GPT-4o-mini with specific instructions to detect known scheduling flaws.
- 3) **Response Parsing:** The JSON response is validated and transformed into typed `FlawDetection` objects.
- 4) **UI Rendering:** Detected flaws are displayed with severity indicators, affected processes, and actionable recommendations.

The analysis detects nine flaw categories:

- Convoy effect, starvation, excessive context switches
- High waiting time, priority issues, quantum inefficiency
- Queue starvation (MLQ/MLFQ), excessive demotion (MLFQ), boost dependency (MLFQ)

IV. RESULTS AND DISCUSSION

A. Experimental Setup

We evaluated all eight algorithms using a benchmark workload of five processes with varying arrival times, CPU burst times, I/O burst times, and priority levels. Table II shows the test workload.

TABLE II
TEST WORKLOAD CONFIGURATION

Process	Arrival	CPU Burst	I/O Burst	Priority
P1	0	8	2	3
P2	1	4	1	1
P3	2	9	3	4
P4	3	5	0	2
P5	4	2	1	5

B. Performance Comparison

Table III presents the key metrics across algorithms.

TABLE III
ALGORITHM PERFORMANCE COMPARISON

Algorithm	Avg Wait (ms)	Avg TAT (ms)	Total Time (ms)	CPU Util. (%)
FCFS	12.4	20.0	29	100
SJF	7.2	14.8	29	96.6
SRTF	5.8	13.4	30	90
Priority	9.6	17.2	29	94
Priority-P	8.4	16.0	30	95
Aging	7.8	15.4	30	97
RR ($q=2$)	10.8	18.4	30	90
MLQ	8.4	16.0	31	96
MLFQ	6.6	14.2	31	98

Note: CPU utilization is computed as $U = \frac{\sum B_i}{T_{\text{total}}} \times 100\%$, where $\sum B_i = 28$ ms (total CPU burst across all processes) and T_{total} is the simulation end time. Because all five processes arrive within the first 4 ms, idle intervals are minimal, yielding utilization in the range of 90–97% across all algorithms.

C. AI Flaw Analysis Results

The flaw analysis engine automatically ran after each simulation and produced the following findings per algorithm. Severity is classified as *high*, *medium*, or *low* based on the fraction of processes affected and the magnitude of the performance impact.

- **FCFS** — *Convoy Effect* (high): P5 (burst = 2 ms) waited 24 ms because it was queued behind P1 (burst = 8 ms) and P3 (burst = 9 ms). Shorter processes are entirely blocked by earlier, longer ones. Recommendation: use SJF or RR for interactive workloads.
- **SJF** — *Starvation* (medium): P3 (burst = 9 ms, the longest job) was consistently deferred in favour of P5 (burst = 2 ms) and P4 (burst = 5 ms), resulting in a turnaround time 58% above the workload average. Recommendation: apply SRTF or introduce aging to guarantee P3 eventually progresses.
- **SRTF** — *Excessive Context Switches* (medium): P3 was preempted 4 times as shorter processes (P4, P5) arrived and reclaimed the CPU. The total context-switch count exceeded $N - 1 = 4$ for five processes, introducing overhead. Recommendation: increase scheduling granularity or switch to RR with a larger quantum for CPU-bound processes.
- **Priority (Non-Preemptive)** — *Priority Issue* (medium): P2 (priority = 1, highest) arrived at $t = 1$ ms but could not preempt P1 (priority = 3) already running, delaying the highest-priority process by 7 ms. Recommendation: switch to preemptive priority scheduling for latency-sensitive processes.
- **Priority (Preemptive)** — *High Waiting Time* (medium): P3 (priority = 4, lowest among arrived processes) experienced a waiting time $2.1 \times$ the workload average because P2, P4, and P5 all pre-empted it repeatedly. Recommendation: add aging to gradually raise P3's priority and cap its waiting time.

- **Priority + Aging** — *Quantum Inefficiency* (low): With the default aging interval $A = 5$, P3's effective priority improved only at $t = 15$ ms. A shorter aging interval would provide earlier relief without causing starvation in higher-priority processes. No starvation was detected; the algorithm performed well overall.
- **Round Robin** ($q = 2$) — *Quantum Inefficiency* (medium): A quantum of 2 ms is small relative to the largest burst of 9 ms (P3), requiring P3 to be rescheduled 5 times. This inflated the context-switch count by 40% compared to FCFS and raised the average waiting time above the SJF baseline. Recommendation: increase the quantum to 4–5 ms to reduce thrashing while preserving fairness.
- **MLQ** — *Queue Starvation* (high): The Background queue (Q3) received zero CPU time during the entire simulation. The System (Q0, FCFS) and Interactive (Q1, RR) queues held the CPU continuously, leaving Background processes indefinitely deferred. This is a structural limitation of fixed-priority MLQ with no inter-queue preemption limit. Recommendation: enforce a time-slice budget per queue level or consider MLFQ for dynamic priority adjustment.
- **MLFQ** — *Boost Dependency* (low) & *Excessive Demotion* (medium): All five processes were eventually demoted to Q2 (lowest queue) after exhausting their Q0 and Q1 quanta, causing MLFQ to behave similarly to FCFS in the latter half of the simulation. The only mechanism restoring fairness was the periodic priority boost at $t = 20$ ms. Reducing the boost interval or increasing Q1's quantum would mitigate this pattern.

V. CONCLUSION

This paper presented an interactive CPU scheduling simulator with AI-powered flaw analysis. The simulator supports eight scheduling algorithms including the complex MLQ and MLFQ variants, and provides automated detection of nine categories of scheduling anomalies. The integration of large language models enables context-aware analysis that goes beyond simple threshold-based checks, offering students and educators actionable insights into algorithm behavior.

Future work includes: (1) extending the analysis to support real-time scheduling algorithms such as EDF and Rate Monotonic, (2) adding multi-core CPU simulation, and (3) enabling collaborative sessions where multiple users can observe and modify the same simulation.

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